

Drone System to Monitor Animals

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The Hawkeye Drone System to Monitor Animals makes use of drones to assist in the tracking and research of animals. This project aims to speed up and simplify wildlife monitoring without requiring constant field presence. The system analyzes the video that is captured by the drone with a camera to identify animals and comprehend the situation. This can be helpful in spotting potential threats like poaching and tracking the movement of animals.

A layered structure connects the drone, backend, and user interface of the system. Through its camera, the drone captures live video and serves as the primary data source. A backend system receives this video and processes it. The backend runs detection models (Yolov26) that try to find animals in each frame. The system keeps track of details like the animal's species and time of detection. A database is used to store this data. A web dashboard lets users see the results on the front end. Users are able to see what the drone is detecting in real time and see live updates on the dashboard.

The system's development took place in stages. Getting the drone camera to work and ensuring that video could be sent to the backend were the first priorities. This step ensured that the system could correctly collect data. Then we trained a Yolov26 model to detect our animals, and did 100 epochs. Following that, a simpler dashboard for displaying results and improved detection accuracy were added to the project. The project evolved from a fundamental concept into a functional system with each step.

The project's testing was an important component. The system was tested several times to ensure that it worked properly. These tests looked for the apps ability to capture video, the backend's ability to process the video, and the accuracy with which animals could be detected. The team also checked to see if the data had been saved correctly and that the information on the dashboard was correct. The majority of the main features worked well, but there are still areas that need work, like detecting animals in dim lighting or difficult conditions.

The system can be enhanced in numerous ways in the future. Improved machine learning models could be used to boost detection accuracy. Also, adding an actual drone would increase mobility so it can cover larger areas. The system could also be expanded to use multiple drones simultaneously. The system would also be better for use in the real world if speed was increased and power consumption was decreased.

Looking back, the project showed how important it is to plan the system structure early. It was simpler to construct and enhance over time by breaking the system into smaller components. Overall, the Hawkeye system is a solid starting point for using technology and drones to monitor animals and clearly has room for expansion.

Proof of Implementation:



